

Structured Data Extraction from MRI Reports Using NLP Techniques: A Comprehensive PhD-Level Literature Review

1. Introduction

The extraction of structured data from MRI reports using natural language processing (NLP) techniques has become a pivotal area in medical informatics, aiming to transform unstructured radiology narratives into actionable, machine-readable information for clinical decision support, research, and workflow optimization [10] [24] [16] [18] [12] [9] [7] [6] [4] [21]. Early efforts focused on rule-based and statistical methods, but recent advances leverage deep learning architectures—especially transformer-based models like BERT and GPT—for tasks such as entity recognition, report classification, and findings extraction [10] [9] [7] [4]. These approaches have demonstrated high accuracy in extracting clinical concepts (e.g., lesions, anatomical locations, imaging features) from free-text MRI reports across various domains including neuroimaging, oncology, and breast imaging [9] [2] [7]. Multimodal fusion—integrating textual data with imaging or clinical metadata—further enhances diagnostic precision and supports personalized medicine [22] [13] [6] [23]. Despite these advances, challenges remain regarding data quality, annotation scarcity, generalizability across institutions, interpretability of deep models, and ethical considerations [24] [12] [19] [5]. This review synthesizes the state-of-the-art in NLP-driven structured data extraction from MRI reports, highlighting methodologies, applications, limitations, and future directions.

Figure 1: Consensus meter visualizing agreement on NLP accuracy for structured data extraction from MRI reports.

2. Methods

A comprehensive search was conducted over 170 million research papers in Consensus—including Semantic Scholar, PubMed, and other databases—using targeted queries related to NLP-based structured data extraction from MRI reports. The search identified 1,055 potentially relevant papers; after de-duplication and screening for relevance to the topic (including foundational theory and adjacent modalities), 418 papers were screened. Of these, 256 met eligibility criteria based on methodological rigor and relevance. The final review included the top 50 most relevant papers.

Identification	Screening	Eligibility	Included
1055	418	256	50

Figure 2: Flow diagram of literature selection process for this review.

Eight unique search groups were executed to ensure comprehensive coverage of foundational methods, technical advances in NLP/AI for radiology/MRI reports, multimodal integration strategies, limitations/barriers to adoption, and adjacent imaging modalities.

3. Results

3.1 Evolution of NLP Techniques for MRI Report Extraction

Early approaches relied on rule-based systems and traditional machine learning (ML) algorithms for information extraction from radiology reports [18] [21]. With the advent of deep learning (DL), convolutional neural networks (CNNs) and recurrent neural networks (RNNs) improved feature representation for both image analysis and text processing [1] [20] [14]. Transformer-based models such as BERT and GPT have recently set new benchmarks in extracting entities and relationships from complex medical narratives [10] [9] [7].

3.2 Applications Across Clinical Domains

NLP-driven structured data extraction has been applied to diverse clinical scenarios:

- **Neuroimaging:** Automated labeling/classification of brain MRI findings for disorders like Alzheimer’s disease or brain tumors [2] [3].
- **Oncology:** Extraction of tumor characteristics from prostate or breast MRI reports to support radiogenomics research [8] [7].
- **Workflow Optimization:** Automated cohort identification for research or quality assurance by parsing large volumes of historical MRI reports [24] [18].

3.3 Multimodal Fusion & Integration

Recent studies emphasize integrating textual information extracted via NLP with imaging features or clinical metadata using multimodal AI frameworks [22] [13] [6]. This fusion enables more robust disease prediction models by leveraging complementary strengths of each modality.

3.4 Challenges & Limitations

Key barriers include:

- **Data Quality & Annotation:** Scarcity of annotated datasets limits supervised model training; domain adaptation remains challenging due to institutional variability in reporting styles [24] [12].
- **Interpretability:** Deep models often function as ”black boxes,” complicating clinical trust and regulatory approval [19].
- **Generalizability:** Models trained on one institution’s data may not perform well elsewhere without retraining or adaptation [5].
- **Ethical & Privacy Concerns:** Handling sensitive patient information requires robust de-identification protocols [19].

Results Timeline results_timeline **Figure 3:** Timeline showing evolution of key methods in NLP-based MRI report extraction. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	Mingzhe Hu	[10] [9]
Author	Sapna Singh Kshatri	[14]
Author	Mahsa Arabahmadi	[2]
Journal	<i>IEEE Reviews in Biomedical Engineering</i>	[24] [17]
Journal	<i>Diagnostics</i>	[19] [7]
Journal	<i>Frontiers in Public Health</i>	[15]

Figure 4: Authors & journals that appeared most frequently in the included papers.

4. Discussion

The literature demonstrates that modern NLP techniques—especially transformer-based models—can achieve high accuracy in extracting structured information from unstructured MRI reports across multiple domains [10] [9] [7]. These advances facilitate secondary uses such as automated cohort identification for research or quality improvement initiatives [18], as well as integration with imaging features for enhanced diagnostic modeling [22] [6]. However, widespread adoption is hindered by challenges related to annotation scarcity (limiting supervised learning), lack of generalizability across institutions due to reporting heterogeneity [5], interpretability concerns with deep models (“black box” issue) [19], and ethical/privacy considerations when handling sensitive patient data.

Despite these challenges, the field is rapidly evolving toward more robust multimodal AI systems that combine textual extractions with image-derived features or clinical metadata for improved predictive performance [22] [13]. There is also growing emphasis on explainable AI (XAI) methods to enhance transparency and clinician trust in automated systems [11].

Claims and Evidence Table

Claim	Evidence Strength	Reasoning	Papers
Transformer-based NLP models can accurately extract entities/relations from MRI reports	Evidence strength: Strong (9/10)	Multiple studies show high F1 scores (≥ 0.85) using BERT/GPT variants on annotated datasets	[10] [9] [7]
Multimodal fusion improves diagnostic/predictive modeling	Evidence strength: Strong (8/10)	Combining text-derived features with imaging/clinical metadata yields better performance than unimodal approaches	[22] [13] [6]
Annotation scarcity limits supervised model development	Evidence strength: Moderate (7/10)	Most studies cite lack of large annotated corpora as a barrier; domain adaptation remains challenging	[24] [12]
Deep learning models face interpretability issues	Evidence strength: Moderate (6/10)	Black-box nature complicates clinical trust/regulatory approval; XAI methods are emerging but underused	[19] [11]
Generalizability across institutions is limited	Evidence strength: Moderate (5/10)	Reporting style heterogeneity reduces out-of-sample performance; few studies address cross-site validation	[5]
Ethical/privacy concerns are insufficiently addressed	Evidence strength: Weak (3/10)	Few studies detail robust de-identification protocols or address privacy risks explicitly	[19]

Figure undefined: Key claims and support evidence identified in these papers.

5. Conclusion

Structured data extraction from MRI reports using advanced NLP techniques has matured significantly over the past decade. Transformer-based models now enable accurate entity/relation extraction at scale; integration with multimodal AI frameworks further enhances diagnostic utility. However, persistent challenges around annotation scarcity, generalizability across sites/institutions, interpretability ("black box"), and privacy/ethics must be addressed before routine clinical deployment.

Research Gaps

Topic/Outcome	Neuroimaging Reports	Oncology Reports	Breast Imaging Reports	Multimodal Fusion Studies
Entity Extraction Accuracy	8	5	4	6
Relation Extraction	5	3	2	5
Cross-Institution Validation	2	1	GAP	1
Explainable AI Integration	1	GAP	GAP	1

Figure undefined: Matrix showing coverage gaps by topic/outcome versus study attribute.

Open Research Questions

Future work should focus on expanding annotated datasets through collaborative efforts; developing domain-adaptive/federated learning strategies; improving model interpretability via XAI; conducting rigorous cross-institutional validations; and establishing robust privacy-preserving protocols.

Question	Why
How can large-scale annotated corpora be developed efficiently for training robust NLP models on MRI reports?	Annotated datasets are essential for supervised learning but remain scarce due to resource constraints; scalable solutions are needed.
What strategies best improve generalizability of NLP models across institutions with heterogeneous reporting styles?	Ensuring consistent performance outside single-institution settings is critical for real-world deployment.
How can explainable AI methods be integrated into NLP pipelines to enhance clinician trust?	Interpretability is vital for regulatory approval and clinician acceptance but remains underexplored in current systems.

Figure undefined: Open questions guiding future research directions.

In summary: While significant progress has been made in automating structured data extraction from MRI reports using advanced NLP techniques—especially transformers—addressing annotation scarcity, generalizability barriers, interpretability issues, and privacy concerns will be crucial for widespread clinical adoption.

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